

Prime nos: number which has exactly 2 factors
 $\downarrow$
 find no. of factors

<u>i</u>	<u>N/i</u>
1	12
2	6
3	4
4	3
6	2
12	1

$$i \leq N/i$$

$$\Rightarrow i^2 \leq N$$

$$\Rightarrow \boxed{i \leq \sqrt{N}}$$

$$i > N/i$$

<u>i</u>	<u>N/i</u>
1	36
2	18
3	12
4	9
6	6
9	4
12	3
18	2
36	1

$\Rightarrow$ checkPrime

boolean isPrime(N) {

 C=0

 if (N==1) return false;

 for (i=2; i<=sqrt(N); i++) {

 if (N%i==0)

 if (i==N/i)

 C++

$$\boxed{\begin{array}{l} TC \Rightarrow O(\sqrt{N}) \\ SC \Rightarrow O(1) \end{array}}$$

```

else
    c = c + 2
}
return c == 2
}

```

$\Rightarrow$ checkPrime(2)

```

boolean isPrime(N) {
    c = 0
    if (N == 1) return false;
    for (i = 2; i <= sqrt(N); i++) {
        if (N % i == 0) {
            return false
        }
    }
    return true
}

```

Q6. Given N, find all the prime nos. between 1 to N.

ex $\Rightarrow$ N = 10 $\Rightarrow$ [1-10] $\Rightarrow$ 2 3 5 7

∴ brute force

try for all nos. (1 to N), if they prime

TC $\Rightarrow O(N\sqrt{N})$ SC $\Rightarrow O(1)$

$N = 50$

1	<u>2</u>	<u>3</u>	4	<u>5</u>	6	<u>7</u>	8	9	10
<u>11</u>	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50

bool $p[N+1] = \text{True}$

$p[0] = \text{False}$; $p[1] = \text{False}$

for ($i = 2$; $i \leq N$; $i++$) {

if ($p[i] == \text{true}$) {

// i is prime

// we need to iterate multiple of $i \leq N$

for ($j = i \times 2$; $j \leq N$; $j = j + i$) {

$p[j] = \text{false}$

}

}

i	j	iteration
2	$N/2$	$N/2$
3		$N/3$
5		$N/5$

this can
be changed to
 $j = i \times i$

can
be
changed
to $\sqrt{N}$

$$TC \Rightarrow N/2 + N/3 + N/5 + \dots + N/p \quad \left[\begin{array}{l} p \text{ is largest prime} \\ \text{no.} \leq N \end{array} \right]$$

$$\Rightarrow N \left[\frac{1}{2} + \frac{1}{3} + \frac{1}{5} + \dots + \frac{1}{p} \right]$$

↓

Sum of reciprocals for prime = $\log(\log N)$

$$TC \Rightarrow N(\log(\log N))$$

$$\Rightarrow 5N$$

$$\approx \underline{\underline{O(N)}}$$

$$N = 2^{31}$$

$$\log_2 N \Rightarrow \log_2 2^{31}$$

$$\Rightarrow 31$$

$$\log_2 31 \Rightarrow 5$$

$$SC \Rightarrow O(N)$$

$$i=2 \rightarrow 2 \times 2 = \checkmark$$

$$i=3 \rightarrow 3 \times 2 = 6 \text{ X}$$

$$3 \times 3 = 9 \checkmark$$

$$i=5 \rightarrow 5 \times 2 = 10 \text{ X}$$

$$5 \times 3 = 15 \text{ X}$$

$$5 \times 4 = 20 \text{ X}$$

$$5 \times 5 = 25 \checkmark$$

$$i=11 \Rightarrow 11 \times 11 = 121$$

$$i \times i \leq N$$

$$\Rightarrow i^2 \leq N$$

$$\Rightarrow i \leq \sqrt{N}$$

$$i=7 \rightarrow 7 \times 2 = 14 \text{ X}$$

$$7 \times 3 = 21 \text{ X}$$

$$7 \times 4 = 28 \text{ X}$$

$$7 \times 5 = 35 \text{ X}$$

$$7 \times 6 = 42 \text{ X}$$

$$7 \times 7 = 49 \checkmark$$

⇒ [Sieve of Eratosthenes]

Q2. Given N , find the smallest prime factor for all nos. $2-N$.

$$ex \Rightarrow 10 \Rightarrow SPF \rightarrow 2$$

$$15 \Rightarrow SPF \rightarrow 3$$

$$17 \Rightarrow SPF \rightarrow 17$$

$$N=10 \Rightarrow$$

	2	3	4	5	6	7	8	9	10
SPF	2	3	2	5	2	7	2	3	2

$$N=25$$

$$idx \Rightarrow 0 \text{ to } 10$$

	2	3	4	5	6	7	8	9	10
SPF	2	3	4 2	5	6 2	7	8 2	9 3	10 2

$$idx = 11 \text{ to } 20$$

	12	13	14	15	16	17	18	19	20
"	12 2	13	14 2	15 3	16 2	17	18 2	19	20 2

$$idx = 21 \text{ to } 25$$

	21	22	23	24	25
	21 3	22 2	23	24 2	25 5

(10)
factor $\rightarrow 1, 2, 5, 10$

multiples $\rightarrow 10, 20, 30$
40, 50, -
- -

spf[N+1] = { initialize spf[i] = i }

for (i=2; i ≤ √N; i++) {
 if (spf[i] == i) ^{→ prime}

for (j=i*i; j ≤ N; j=j+i) {

if (spf[j] == j)

spf[j] = i

}

}

TC = O(N)
 SC = O(N)

Q3. Given a no. find the count of divisors / factors

ex → x = 72 ⇒ 2³ × 3²

no of factors = (3+1) (2+1) = 12

$$\begin{array}{r} 2 \overline{) 72} \\ \underline{2} \\ 36 \\ 2 \overline{) 36} \\ \underline{2} \\ 18 \\ 2 \overline{) 18} \\ \underline{2} \\ 9 \\ 3 \overline{) 9} \\ \underline{3} \\ 0 \end{array}$$

$$\begin{array}{l} 2^3 \rightarrow 2^0 \ 2^1 \ 2^2 \ 2^3 \\ 3^2 \rightarrow 3^0 \ 3^1 \ 3^2 \end{array}$$

2⁰ × 3⁰ ⇒ 1

2¹ × 3⁰ ⇒ 2

2² × 3⁰ ⇒ 4

2³ × 3⁰ ⇒ 8

2⁰ × 3¹ ⇒ 3

2¹ × 3¹ ⇒ 6

2² × 3¹ ⇒ 12

2³ × 3¹ ⇒ 24

2⁰ × 3² ⇒ 9

2¹ × 3² ⇒ 18

2² × 3² ⇒ 36

2³ × 3² ⇒ 72

all are factors of 72 $\Rightarrow$ 72

2^0

1

2^0

2^1

2^2

2^1

1

2^0

2^1

2^2

(x)

$\Rightarrow x+1$

(y)

$\Rightarrow y+1$

}

factors $\Rightarrow (x+1) \times (y+1)$

$$ex \Rightarrow 600 \Rightarrow 2^3 \times 3^1 \times 5^2$$

$$\begin{aligned} \hookrightarrow \text{factor} &= (3+1) \times (1+1) \times (2+1) \\ &= 24 \end{aligned}$$

$$\begin{array}{r} 2 \overline{) 600} \\ 2 \overline{) 300} \\ 2 \overline{) 150} \\ 3 \overline{) 75} \\ 5 \overline{) 25} \\ 5 \overline{) 5} \\ 1 \end{array}$$

generalise $\Rightarrow N$ (p_1, p_2, p_3 — are prime)
factors of N

$$N = p_1^x + p_2^y + p_3^z + \dots$$

$$\begin{array}{l} \text{no. of} \\ \text{factors} \end{array} = (x+1) \times (y+1) \times (z+1)$$

$$2 \overline{) 360}$$

$$2 \overline{) 180}$$

$$2 \overline{) 90}$$

$$3 \overline{) 45}$$

$$3 \overline{) 15}$$

$$5 \overline{) 5}$$

1

$$2 - 1 \rightarrow (2-1)$$

$$3 - 1 \rightarrow (3-1)$$

$$5 - 1 \rightarrow (5-1)$$

$$(2+1) \times (3+1) \times (5+1)$$

$$= 4 \times 3 \times 2$$

$$= 24$$

$$\tau(N) = O(\log N)$$

⇒ Find total factors of N.

$$\text{total} = 1$$

while (N > 1) {

$$p = \text{spf}[N]$$

$$c = 0$$

while (N % p == 0) {

$$c = c + 1$$

$$N = N / p$$

}

$$\text{total} = \text{total} \times (c + 1)$$

}

⇒ Find prime factorization of N

list <int> fact;

while (N > 1) {

$$p = \text{spf}(N)$$

fact.add(p)

$$N = N / p$$

}

b) find no. of factors for all nos 1 to N

Step 1 $\rightarrow$ create spf array $\rightarrow O(N)$

POD
2 $\rightarrow$ for each value of N, find
no. of factors $\rightarrow O(\log N)$

$$\boxed{\begin{array}{l} TC \approx O(N \log N) \\ SC \approx O(N) \end{array}}$$

Q 4. Are primes finite or infinite?

assume

finite prime $\Rightarrow p_1, p_2, p_3, p_4, p_5, \dots, p_N$

$\Rightarrow$ multiply all the primes

consecutive $\left[\begin{array}{l} x = p_1 \times p_2 \times p_3 \times p_4 \times \dots \times p_N \\ y = x + 1 \end{array} \right.$

$$\Rightarrow \boxed{\text{gcd of any two consecutive nos.} = 1}$$

$$\gcd(x, y) = 1.$$

↓

they have no common divisor
between x & y

p_1
 p_2
 p_3
 $\vdots$
 p_n

→ none of them are factors of y .

that means y is also a prime no.

[CONTRADICTION]

Hence proved primes are infinite